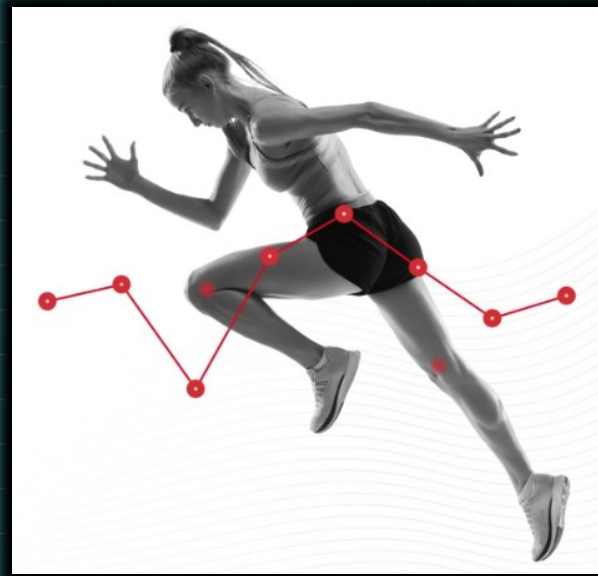


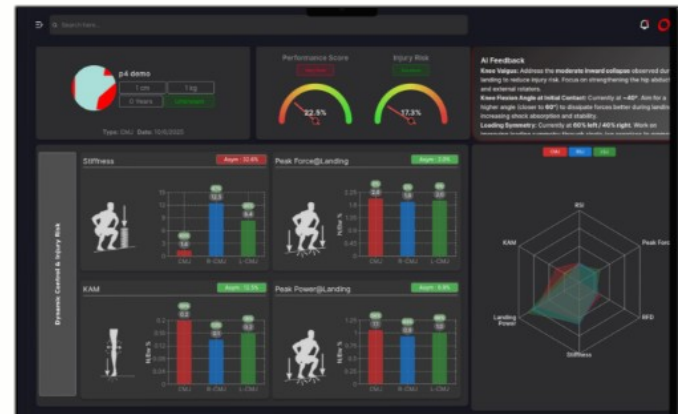
OptiZone – Semester Project



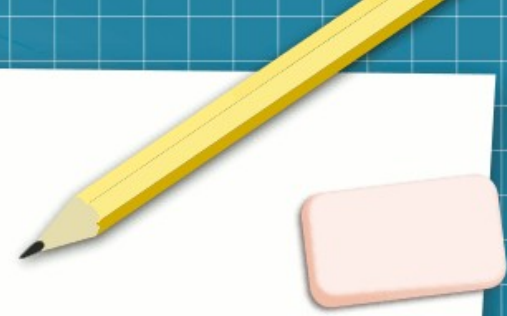
presented by: Michel ABELA

Project Motivation

- in the past 15 years:
 - Increase in physical demand for modern soccer
 - Need for **injury prevention (ACL)** & **performance optimisation**
 - Need for athlete **workload monitoring & visualisation**
- **Wearable technology** enables data collection (**GPS & HR**)



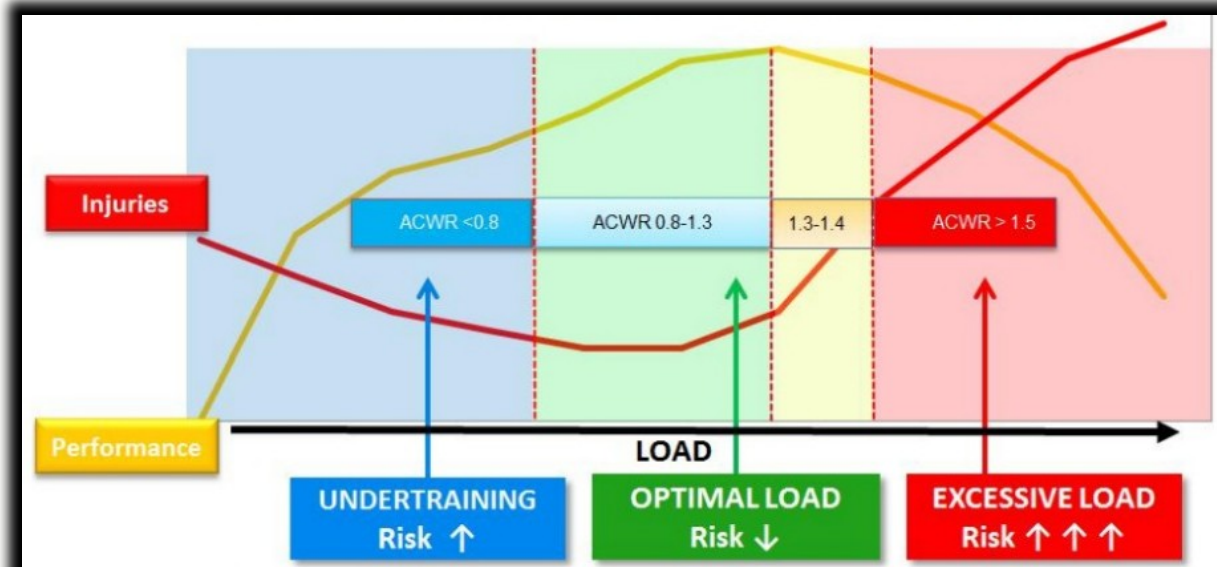
How to use GPS & HR data ?



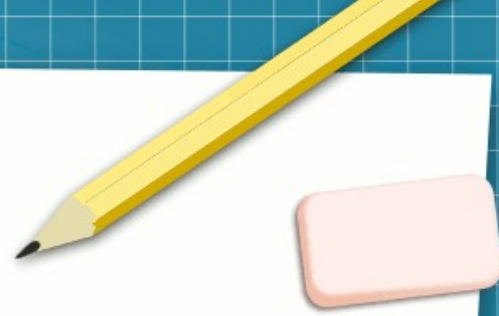
- Determine workload management:
 - **External Load:**
 - Athlete's physical load
 - What the athlete does during the activity
 - **GPS** based metrics (*e.g. distance – speed – acceleration*)
 - **Internal Load:**
 - Athlete's physiological load
 - How the athlete responds to external load during the activity
 - **Heart Rate** based metrics (*HR zones – TRIMP – Recovery Rate*)
- **Excessive fatigue** → higher risk of anterior cruciate ligament (ACL) injury

Optimal Workload Management

- Acute/Chronic Workload Ratio (ACWR):
 - Measures relation between current week load & average week load
- Relationship: Load – Performance - Injury

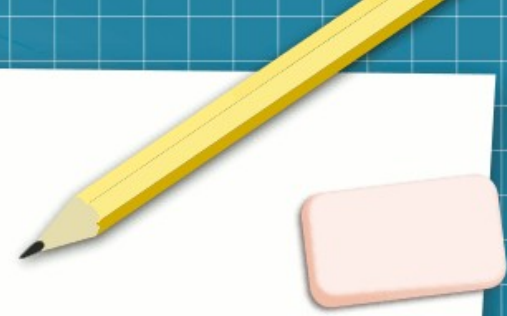


Key GPS workload metrics



- Basic Movement metrics:
 - **Total distance**
 - **Speed/Pace** – *instantaneous, average, max*
 - **Acceleration/Deceleration** – *rate of speed change*
 - **Moving/Stopped time**
- Segmented Performance metrics:
 - **Split Pace** – *average pace per Km*
 - **Time in Speed Zones** – *time percentage spent in speed zones*
 - **Best Segment time** – *max pace among split paces*

Key HR workload metrics



- Cardiovascular Load metrics:
 - **Average/Max Heart Rate**
 - **HR Reserve** – *range of reserve for max effort*
 - **Time in HR Zones** – *time spent in %HRmax zones*
 - **Training Impulse (TRIMP)** – *weighted measure cardiovascular load*
- Fitness Indicator metrics:
 - **Pace v/s HR** – *indicates aerobic efficiency*
 - **Running Economy** – *pace per heartbeat*
 - **Cardiac Cost** – *HR per unit distance*

Why visualisation matters ?

- **Athletes & Coaches** need fast, intuitive insights
- **Graphs** highlight trends & anomalies to **improve training**

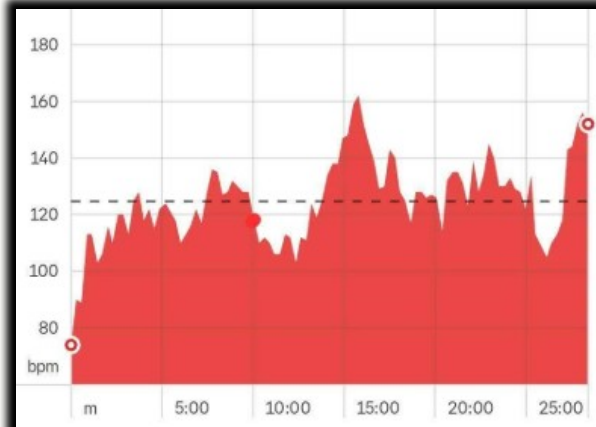
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Time	Lat	Lon	Speed (m/s)	Heart Rate (bpm)	Hacc	Hdop	Quality of Signal	No. of Satellites	Instantaneous Acceleration Impulse	Accel X	Accel Y	Accel Z	Gyro Yto X	Gyro Y	Gyro Z
2	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	0.162504	-0.178608	0.016104	24.22	-77.77	41.93
3	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	0.084912	-0.10248	-0.004392	23.03	-71.4	30.24
4	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.041724	-0.077592	-0.041724	22.96	-60.34	18.34
5	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.15006	-0.087108	-0.050508	21.91	-47.32	12.67
6	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.20496	-0.115656	-0.016104	22.33	-39.69	14.84
7	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.207156	-0.149328	0.020496	24.43	-47.46	20.79
8	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.169824	-0.184464	0.016104	24.57	-64.33	27.09
9	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.099552	-0.200568	-0.01098	21.28	-74.48	32.41
10	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	-0.024888	-0.17568	-0.026352	16.94	-77.07	35.28
11	08:39:19.1	46.2198695	6.082511	1.38333444	140	0	0.4	337	19	0.3055558000000001	0.03294	-0.13176	-0.03294	12.39	-78.82	35.91
12	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.074664	-0.092232	-0.030744	8.54	-78.05	36.19	
13	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.099552	-0.061488	-0.016836	5.81	-75.88	35.98	
14	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.109068	-0.037332	-0.005856	3.36	-77.07	35.49	
15	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.098088	-0.01464	-0.012444	0.56	-81.55	35.42	
16	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.059292	0.008784	-0.0366	-2.52	-85.26	34.02	
17	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	0.00732	0.022692	-0.068808	-5.39	-85.4	30.59	
18	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	-0.046116	0.019764	-0.098088	-8.33	-82.81	26.53	
19	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	-0.095892	0.001464	-0.114192	-12.39	-80.15	23.03	
20	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	-0.13542	-0.032208	-0.105408	-16.87	-76.65	21.63	
21	08:39:19.2	46.2198675	6.08251083333333	2.1944462	140	0	0.4	337	19	-0.16104	-0.06954	-0.072468	-21.42	-74.97	22.96	
22	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.165432	-0.088572	-0.020496	-23.03	-71.68	26.11
23	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.140544	-0.079788	0.019032	-18.13	-64.26	31.08
24	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.097356	-0.057096	0.016836	-12.18	-56.28	36.47
25	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.064416	-0.061488	0	-9.38	-51.45	39.76
26	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.054168	-0.0915	-0.01098	-8.96	-47.67	41.37
27	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.065148	-0.095892	-0.010248	-9.1	-42.42	42.91
28	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.079056	-0.06222	0.0366	-7.63	-38.22	44.59
29	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.071736	-0.013176	0.154452	-2.31	-40.25	46.62
30	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	0.016836	0.1464	0.794952	12.39	-56.42	48.3
31	08:39:19.3	46.2198656666667	6.08251033333333	2.17777952	140	0	0.4	337	19	0.1666667999999997	-0.074664	0.72834	1.595028	75.18	-96.53	45.5
32	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-0.99552	1.9215	1.41276	112.35	84.98	35.77
33	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-1.717272	3.232512	1.638216	135.8	403.83	-6.02
34	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-2.094984	3.987936	2.805756	176.4	111.16	-71.75
35	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-2.125728	3.08904	2.468304	176.4	-65.17	-69.51
36	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-0.892308	1.144848	1.307352	187.04	-0.14	-98.56
37	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	0.411384	-0.131028	1.444968	87.15	-91.14	-112.77
38	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	0.821304	-0.494832	1.911984	60.69	-111.65	-85.26
39	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	0.652944	-0.204228	1.439844	130.13	-97.72	-51.87
40	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	0.357216	0.272304	1.05774	85.12	-38.85	-29.12
41	08:39:19.4	46.219864	6.08251	1.85833482	140	0	0.4	0	0	3.194447	-0.024156	0.686616	1.35786	58.59	13.3	-8.82
42	08:39:19.5	46.2198625	6.0825095	1.60833462	140	0	0.4	337	19	2.500002	-0.5124	0.908412	1.540128	90.79	17.36	-2.59

Visualisation Examples

- Pace v/s distance:



- Heart Rate v/s Time:



Project Code implementation

- Python professional backend
 - *Built separate modules as libraries (API structure)*
- CSV Dataset handler
- Workload Metrics calculator
- Visualisation Plots handler
- Parameters Configuration
- PDF Report generator

